

Lesson 6



Fixed X , $y = X\beta + u$

$$H_0: R\beta = r$$

$$q \times 1 \quad k \times 1 \quad q \times 1$$

$$\frac{1}{\sigma^2} (Rb - r)' [R(X'X)^{-1}R']^{-1} (Rb - r) \sim \chi^2_q$$

$$\frac{e'e}{n-k} = s^2 \Rightarrow \frac{e'e}{n-k} \quad \frac{1}{\sigma^2} u'Mu \sim \chi^2_{n-k}$$

$$F(q, n-k) = \frac{(Rb - r)' (R(X'X)^{-1}R')^{-1} (Rb - r) / q}{e'e / n-k}$$

$$y = X_1\beta_1 + X_2\beta_2 + u$$

$$H_0: \beta_1 = 0$$

$$\left[\begin{array}{c|c} I_{k_1} & 0 \\ \hline & K_1 \times K_2 \end{array} \right] \left. \vphantom{\begin{array}{c|c} I_{k_1} & 0 \\ \hline & K_1 \times K_2 \end{array}} \right\} k_1$$

$$R\beta = \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \end{array} \right) \begin{pmatrix} \beta_1 \\ \vdots \\ \beta_k \end{pmatrix}$$

$$b_1 = (X_1' M_2 X_1)^{-1} X_1' M_2 y$$

$$b_1 \sim N(0, \sigma^2 (X_1' M_2 X_1)^{-1})$$

$$F(K, n-K) = \frac{b_1' [X_1' M_2 X_1] b_1 / K_1}{e'e / n - K}$$

$$y = X_1 b_1 + X_2 b_2 + u y$$

$$M_2 y = M_2 X_1 b_1 + \underbrace{M_2 u y}_{u y} \Rightarrow y' M_2 y = b_1' X_1' M_2 X_1 b_1 + y' u y$$

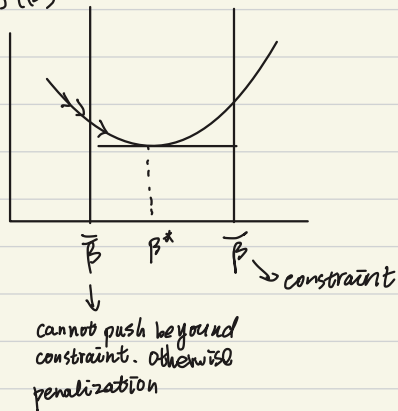
$$b_1' X_1' M_2 X_1 b_1 = y' M_2 y - y' u y$$

$$F(K_1, n-K) = \frac{(y' M_2 y - y' u y) K_1}{y' u y / n - K}$$

$$M_2 y \quad u y \quad H_0: \beta = 0$$

$$F(K, n-K) = \frac{(RSSR - USSR) / K_1}{USSR / n - K}$$

Digression $f(\beta)$



$$\beta < \bar{\beta}$$

$$\min_{b^*, \lambda^*} L = (y - Xb^*)'(y - Xb^*) - 2(Rb^* - r)' \lambda^*$$

$\begin{matrix} k \times q & & q \times 1 \\ \swarrow & & \searrow \\ & \text{estimate of } \lambda^* & \end{matrix}$

$\min_{b^*, \lambda^*} e_*' e_*$
 s.t. $Rb^* = r$

$$y'y - b_*' X'y - y' X b_* + b_*' X' X b_* - (Rb_* - r)'$$

$$FOC = -2X'y + 2X'Xb_* + 2R'\lambda_* = 0$$

$$Rb_* - r = 0$$

$$\rightarrow -X'y + X'Xb_* + 2R'\lambda_* = 0$$

$$\begin{aligned} -R(X'X)^{-1}X'y + R(X'X)^{-1}X'Xb_* &= R(X'X)^{-1}R'\lambda_* \\ -Rb + Rb_* &= (R(X'X)^{-1}R')\lambda_* \\ -Rb + r &= R(X'X)^{-1}R'\lambda_* \\ \lambda_* &= (R(X'X)^{-1}R')^{-1}(r - Rb) \end{aligned}$$

Substitute back λ^*

$$-X'y + X'Xb^* = R'[R(X'X)^{-1}R']^{-1}(r - Rb)$$

$$-(X'X)^{-1}X'y + \underset{\downarrow}{b}^* = (X'X)^{-1}R'[R(X'X)^{-1}R']^{-1}(r - Rb)$$

$$\begin{aligned} b^* &= b + (X'X)^{-1}R'[R(X'X)^{-1}R']^{-1}(r - Rb) \\ &= b - (X'X)^{-1}R'[R(X'X)^{-1}R']^{-1}(Rb - r) \end{aligned}$$

$$e_x = y - Xb^* + Xb - Xb$$

$$= y - Xb - X(b^* - b)$$

$$= e - X(b^* - b)$$

$$e_x'e_x = e'e + (b^* - b)'X'X(b^* - b)$$

$$F(q, n-k) = \frac{(Rb - r)' [R(X'X)^{-1}R']^{-1} (Rb - r)}{e'e / n - k} / q$$

$$e_*' e_* = (r - Rb)' [R(X'X)^{-1}R']^{-1} R(X'X)^{-1} X'X (X'X)^{-1} R' [R(X'X)^{-1}R']^{-1} (r - Rb)$$

$$= (Rb - r)' [R(X'X)^{-1}R']^{-1} Rb - r$$

$$\text{then } F(q, n-k) = \frac{(e_*' e_* - e'e) / q}{e'e / n - k} = \frac{(RSSR - USSR) / q}{USSR / n - k}$$

$$b* = b + \underbrace{(x'x)^{-1} R' [R(x'x)^{-1} R']^{-1}}_A (Rb - r)$$

$$\beta + (x'x)^{-1} x'u - AR(\beta + (x'x)^{-1} x'u) + Ar$$

$$= \beta + (x'x)^{-1} x'u - AR\beta - AR(x'x)^{-1} x'u + Ar$$

$$E(b*) = \beta - AR\beta + Ar \quad \text{Hunder } Rb = r$$

$$= \beta \quad \text{if } H_0 \vee \text{ otherwise biased.}$$

$$\text{Var}(b*) = E[(b* - \beta)(b* - \beta)']$$

$$= E\left((x'x)^{-1} x'u - AR(x'x)^{-1} x'u\right) \left(u'x(x'x)^{-1} - u'(x'x)^{-1} R' A'\right)$$

$$= E\left((x'x)^{-1} x'u u' x (x'x)^{-1} - (x'x)^{-1} x'u u' x (x'x)^{-1} R' A' - AR(x'x)^{-1} x'u u' x (x'x)^{-1} + AR(x'x)^{-1} x'u u' x (x'x)^{-1} R' A'\right)$$

$$= \underbrace{G^2 (X'X)^{-1}}_{V(b)} - \underbrace{G^2 (X'X)^{-1} R' A'}_{\text{①}} - \underbrace{G^2 A R (X'X)^{-1}}_{\text{②}} + \underbrace{G^2 A R (X'X)^{-1} R' A'}_{\text{③}}$$

① and ② and ③ are the same

$$\text{Var}(b^*) = \text{Var}(b) - \text{p.d. matrix}$$

Even more efficient than b

$$b^* = \beta - AR\beta + Ar + (X'X)^{-1} X'U - AR(X'X)^{-1} X'U$$

$$\text{Var}(b^*) = E[(b^* - E(b^*))(b^* - E(b^*))']$$

↳ Claude:

Add discussion on when b^* biased and unbiased